

Determining Quadrupole Deformation β_2 Using Coulomb Excitation

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Theory Quadrupole Moments

$$R(\theta, \phi) = R_{\text{average}} \left[1 + \beta Y_{20}(\theta, \phi) \right]$$

$$Q_0 = \frac{3}{\sqrt{5\pi}} Z R_0^2 \beta_2$$

$$B(E2) = \frac{5}{16\pi} e^2 Q_0^2$$

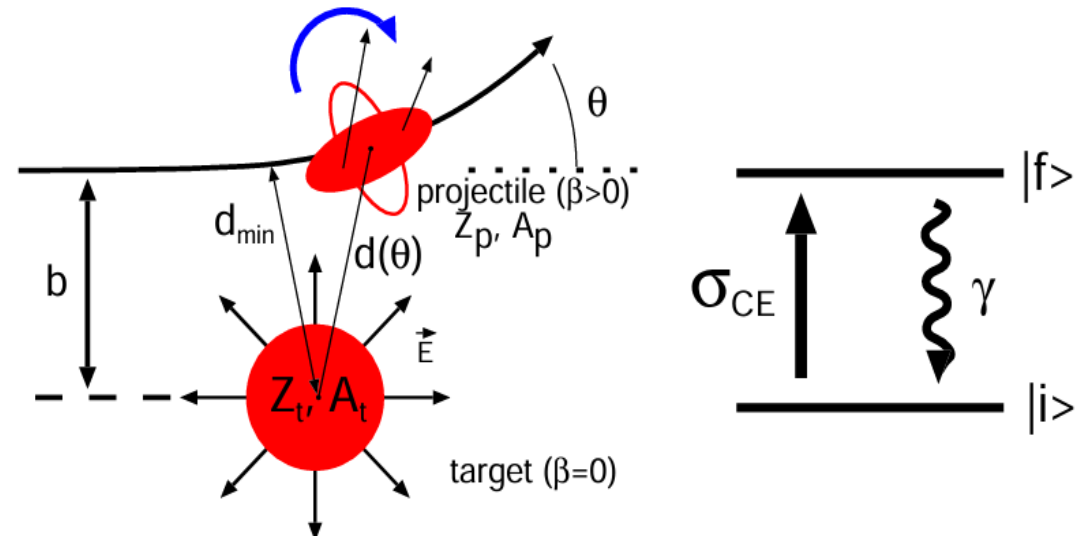
$$\beta_2 = \frac{4\pi}{3ZeR_0^2} \sqrt{B(E2)}$$

- Measuring $B(E2)$ gives β_2 directly.

- Quadrupole Moments first needed to be described by the collective expansion of how a nuclear surface is defined.
- Along with the estimate for reduced transition probabilities, the quadrupole deformation parameter can be derived. Usually, the first attempt
- Key IDEA!!!: needing to use experimental and analytical steps to obtain reduced transition probabilities.

Coulomb Excitation Basics

- Bombard nucleus with electromagnetic field (below Coulomb barrier)
- Pure EM interaction: no nuclear force.
- Observe scattering + γ rays.



Experimental Techniques

- Inverse kinematics with RIB.
- High- Z target (Pb/Sn).
- Silicon detectors + HPGe (e.g. MINIBALL).
- Measure γ yields vs. angle.

* Need to add figures of detectors of interest. This slide will address how these detectors and experiments make important measurements. Essentially, how they work and what processes start the analysis stages. Overall, could be about 2-3 slides

Methodology for Extracting β_2

- ① Extract γ peaks and correct efficiencies.
- ② Normalize using target excitation or elastic scattering.
- ③ Fit data with GOSIA to obtain E2 matrix elements.
- ④ Compute $B(E2)$ and convert to β_2 .

* Each step will be made into its own slide with figures

Result Examples of Literature

References

- [1] Aeshah Ali Hussein et al. “Calculation of the quadrupole deformation parameter (β_2) and reduced transition probability $B(E2; \uparrow)$ of some even-even semi-magic nuclei”. In: *AIP Conference Proceedings*. Vol. 3036. 2024, p. 050007. doi: 10.1063/5.0196266.
- [2] L. Morrison, K. Hadynska-Klepek, Zs. Podolyak, et al. “Quadrupole deformation of ^{130}Xe measured in a Coulomb-excitation experiment”. In: *Physical Review C* 102.5 (2020), p. 054304. doi: 10.1103/PhysRevC.102.054304.
- [3] E. Illana et al. “Low-energy Coulomb excitation of neutron-rich $^{74,76}\text{Zn}$ isotopes”. In: *Physical Review C* 108.4 (2023), p. 044305. doi: 10.1103/PhysRevC.108.044305.
- [4] Oliver Niedermaier. “Coulomb excitation of neutron-rich magnesium isotopes”. PhD thesis. University of Heidelberg, 2005.
- [5] David Verney. “History of the concept of nuclear shape”. In: *European Physical Journal A* 61 (2025), p. 82. doi: 10.1140/epja/s10050-025-01545-1.
- [6] M. Zielińska, J. Srebrny, K. Hadynska-Klepek, et al. “Analysis methods for safe Coulomb excitation with radioactive ion beams using GOSIA”. In: *European Physical Journal A* 52 (2016), p. 99. doi: 10.1140/epja/i2016-16099-8.

- Slide for Result 30/32 Mg B(E2)
- Result example for $^{74/76}\text{Zn}$ B(E2)
- Slide that has overview information and interpretations from results



Conclusions

- Coulomb excitation gives direct access to $B(E2)$.
- From $B(E2) \rightarrow \beta_2$.
- Technique is powerful for both stable and exotic nuclei.